

## Dataset A320

### Geometric data

|                 | Description                                    | Symbol           | Value   | Unit  |
|-----------------|------------------------------------------------|------------------|---------|-------|
| Wing            | Wing reference area                            | S                | 122.40  | $m^2$ |
|                 | Wingspan                                       | b                | 34.100  | m     |
|                 | Aspect ratio                                   | AR               | 9.3900  | -     |
|                 | 25% sweep angle                                | $\Lambda_{25\%}$ | 0.4356  | rad   |
|                 | Root chord                                     | $C_{root}$       | 6.0700  | m     |
|                 | Kink chord                                     | $C_{kink}$       | 3.7600  | m     |
|                 | Tip chord                                      | $C_{tip}$        | 1.5000  | m     |
|                 | Mean aerodynamic chord                         | $C_{MAC}$        | 4.1935  | m     |
|                 | Root relative thickness                        | $t/C_{root}$     | 0.1515  | -     |
|                 | Kink relative thickness                        | $t/C_{kink}$     | 0.1175  | -     |
|                 | Tip relative thickness                         | $t/C_{tip}$      | 0.1084  | -     |
|                 | Oswald factor (clean configuration)            | $e_{clean}$      | 0.8500  | -     |
|                 | Oswald factor (take-off configuration)         | $e_{TO}$         | 0.8000  | -     |
|                 | Oswald factor (approach/landing configuration) | $e_L$            | 0.7500  | -     |
|                 | Taper ratio                                    | $\lambda$        | 0.2471  | -     |
|                 | % location of max. thickness                   | $x_{max\_t}$     | 39.900  | %     |
|                 | $\alpha(C_L=0)$                                | $\alpha_{oL}$    | -0.0524 | rad   |
|                 | $\alpha(C_L=max)$                              | $\alpha_{oL}^*$  | 0.1396  | rad   |
|                 | Leading Edge Radius                            | $r_{LE}$         | 0.0200  | -     |
|                 | Root wing incidence (to fuselage)              | $i_W$            | 0.0639  | rad   |
|                 | Distance from nose to wing apex                | $x_{0,W}$        | 11.760  | m     |
|                 | Wing airfoil type (assumption)                 | NACA 64A410      |         |       |
| Fuselage        | Total length                                   | l                | 37.5700 | m     |
|                 | Height                                         | H                | 4.1400  | m     |
|                 | Width                                          | w                | 3.9500  | m     |
| Horizontal tail | Area                                           | $S_{HT}$         | 31.0000 | $m^2$ |
|                 | Span                                           | $b_{HT}$         | 12.4500 | m     |
|                 | Aspect ratio                                   | $AR_{HT}$        | 5.0000  | -     |
|                 | Taper ratio                                    | $\lambda_{HT}$   | 0.2560  | -     |
|                 | 25% sweep angle                                | $\Lambda_{25HT}$ | 0.5061  | rad   |
|                 | Airfoil rel. thickness                         | $t/C_{HT}$       | 0.1000  | -     |
|                 | Root chord                                     | $C_{rootHT}$     | 3.7400  | m     |
|                 | Distance from nose to horizontal tail apex     | $x_{0,HT}$       | 31.5500 | m     |

|                      | <b>Description</b>     | <b>Symbol</b>    | <b>Value</b> | <b>Unit</b> |
|----------------------|------------------------|------------------|--------------|-------------|
| <b>Vertical tail</b> | Area                   | $S_{VT}$         | 21.5000      | $m^2$       |
|                      | Span                   | $b_{VT}$         | 12.5200      | $m$         |
|                      | Aspect ratio           | $AR_{VT}$        | 1.8200       | -           |
|                      | Taper ratio            | $\lambda_{VT}$   | 0.3030       | -           |
|                      | 25% sweep angle        | $\Lambda_{25VT}$ | 0.5934       | $rad$       |
|                      | Airfoil rel. thickness | $t/c_{VT}$       | 0.1000       | -           |
|                      | Root chord             | $c_{rootVT}$     | 5.4300       | $m$         |

### Propulsion data

|  |                    |           |        |     |
|--|--------------------|-----------|--------|-----|
|  | Unit engine thrust | $T_{eng}$ | 118000 | $N$ |
|  | Engine length      | $l_{eng}$ | 4.44   | $m$ |
|  | Engine width       | $w_{eng}$ | 2.37   | $m$ |

### Mass data

|  |                        |              |       |      |
|--|------------------------|--------------|-------|------|
|  | Maximum take-off mass  | $m_{TO,max}$ | 73500 | $kg$ |
|  | Maximum landing mass   | $m_{L,max}$  | 64500 | $kg$ |
|  | Maximum zero-fuel mass | $m_{ZF,max}$ | 61000 | $kg$ |
|  | Unit engine mass       | $m_{eng}$    | 2366  | $kg$ |

### Unit Conversion Factors

|          |                                            |
|----------|--------------------------------------------|
| Length   | 1 m = 3.2808 ft                            |
| Area     | 1 m <sup>2</sup> = 10.7639 ft <sup>2</sup> |
| Mass     | 1 kg = 2.2046 lbm                          |
| Force    | 1 N = 0.2248 lbf                           |
| Velocity | 1 m/s = 1.944 kn                           |

## Three view A320

